

Phaneendra Ganji

Data Scientist – AI/ML

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Summary

Data Scientist with **over 4** years of experience designing and deploying machine learning systems that solve real business problems and improve operational outcomes. Skilled in the end-to-end lifecycle of AI projects, including data preparation, feature engineering, model development, and performance evaluation using Python, SQL, and frameworks such as TensorFlow and PyTorch. Strong software engineering foundation with practical experience in building APIs, designing system architectures, and developing cloud-based applications on AWS and GCP. Manage model versioning, containerization, and deployment workflows to ensure consistent, reliable, and monitorable production environments.

Skills

Languages: Python, SQL, R, Java, T-SQL
GenAI & LLMs: LangChain, LlamaIndex, Hugging Face Transformers, OpenAI API, LLM Fine-tuning (LoRA, QLoRA, PEFT), Prompt Engineering, RAG, Agentic AI
Database: MySQL, PostgreSQL, MongoDB, DynamoDB, Cosmos DB
Vector Databases: Pinecone, Chroma, Weaviate, Embedding Generation, Semantic Search, Vector Similarity Search
ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, CNNs, LSTMs, Transformers, Autoencoders, Model Evaluation, Bias Analysis
MLOps & DevOps: MLflow, Apache Airflow, Docker, Kubernetes, CI/CD (Jenkins), GitHub, Drift Detection, RESTful APIs, FastAPI, Flask
Cloud Platforms: Azure (ADK, SQL, Blob Storage, AKS), AWS (SageMaker, EC2, Lambda, S3, API Gateway), GCP
Data Engineering: Pandas, NumPy, ETL, Feature Engineering, Data Quality Validation, Apache Spark
Visualization: Power BI, Tableau, Matplotlib, Seaborn, Streamlit
Collaboration: Agile / Scrum, JIRA, Cross-functional Stakeholder Communication, Technical Documentation

Work Experience

ServiceNow, USA July 2024 – Present

Data Scientist (AI/ML Engineer)

- Designed and maintained **PySpark data pipelines** that processed **2,000+ daily platform** events into feature stores, reducing data preparation time by 40% through streamlined ingestion and transformation logic.
- Implemented **classification models (Random Forest)** for automated ticket routing, improving assignment accuracy, and developed **time-series forecasts (Prophet/ARIMA)** to predict incident volumes across 15+ ServiceNow instances, with MAPE within business-accepted thresholds.
- Fine-tuned **BERT-based transformers** for intent classification and built **retrieval-augmented generation (RAG)** pipelines using OpenSearch vector search to retrieve the top 5 relevant knowledge articles, which reduced resolution time.
- Engineered **dynamic prompt templates** for large language models (**GPT/LLaMA**) to generate problem summaries from short ticket descriptions achieved 70% user acceptance in an **A/B test** and saved agents approximately 3 minutes per ticket.
- Containerized **6 models using FastAPI and deployed on Kubernetes (EKS)**, handling 5,000 requests per minute with latency under 200ms; automated CI/CD through GitHub Actions, which lowered deployment rollback frequency.
- Deployed **inference services on AWS EC2 and Lambda** to support up to 150 concurrent users, and maintained an OpenSearch vector index with 1,000+ embeddings, ensuring index refresh completed within 15 seconds.
- Led **3 A/B tests comparing model variants (clustering vs. fine-tuned)** across over 20,000 tickets, documented annual savings of \$0.7M through reduced manual triage, and presented findings quarterly to ServiceNow product leadership.

Neon IT System, India January 2020 – July 2022

Data Scientist (ML Engineer)

- Designed and deployed **predictive maintenance** and **network analytics** models using **Python, TensorFlow, PyTorch, and Scikit-learn**, contributing to increased system uptime and reduced infrastructure failures.
- Built document processing and chatbot automation solutions using **Hugging Face** and **LangChain**, with **generative AI components** to handle unstructured text and support internal workflows.
- Developed and maintained **data ingestion** and **transformation pipelines** with **Pandas, NumPy, and Apache Spark**, ensuring consistent data quality and feature readiness for large-scale telecom and IT datasets.
- Managed **end-to-end MLOps workflows** including model versioning, automated retraining, and monitoring using **Docker, MLflow, GitHub Actions, and Jenkins**, with cloud deployment on **Google Cloud Platform (BigQuery, Cloud Run, Cloud Storage)**.
- Created **RESTful and GraphQL APIs using FastAPI and Flask** to serve models as microservices, enabling seamless integration with front-end and backend systems.
- Automated repetitive business processes and data validation tasks using **Pytest** and **LangGraph**, reducing manual effort and improving operational efficiency.
- Implemented **vector-based search** and recommendation systems using **Pinecone**, and **Chroma**, and integrated **MongoDB** for structured document storage and hybrid retrieval workflows.

Education

Master of Science in Data Science & Analytics,
Florida Atlantic University, FL, USA May 2024

P.G. Diploma in Machine Learning & Artificial Intelligence,
IIT Bangalore, India Aug 2021

Bachelor of Engineering in Computer Science,
Sphoorthy Engineering College, India July 2020